

Ritabrata Patra

Kolkata, West Bengal

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EDUCATION

VIT Bhopal University Bachelor's Degree, Computer Science	2023 – 2027 GPA: 7.91/10.0
National Gems School Class 12th, ICSE Board	2023 81%
National Gems School Class 10th, ICSE Board	2021 93.6%

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL
- **AI / Machine Learning:** Scikit-learn, Random Forest, Logistic Regression, OpenCV, NLP Basics, Model Evaluation
- **Full Stack Development:** HTML5, CSS3, JavaScript, React.js, Flask
- **Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook
- **Core Concepts:** Data Structures (DSA), Problem Solving, Product Thinking, Research & Analysis, Team Leadership, Communication

EXPERIENCE

Prodigal AI	Website Audit & Optimization Intern
• Conducted website audits covering SEO, content, accessibility, and performance. • Performed web scraping and brand research to extract and analyze website data.	

PROJECTS

Malware Detection using Machine Learning	<i>Group Project – Major Contributor</i>
• Built a cybersecurity tool to detect malware in PE/.bat files and classify URLs as safe or malicious. • Implemented Random Forest with 95% accuracy for PE analysis, heuristic rules for batch files, and Logistic Regression with NLP achieving 88% accuracy for URL classification. • Designed a Tkinter GUI for real-time results, demonstrating applied AI in a practical security context.	
Digital Document Signing Application	<i>Independent Project</i>
• Developed a zero-dependency Java desktop application for digitally signing and managing documents. • Built a freehand signature canvas, PDF import with metadata extraction, signer detail capture, and signed HTML export. • Designed the application as a single-file Java Swing solution without external libraries.	
Blockchain-based Face Recognition Voting System	<i>Group Project – Major Contributor</i>
• Built a secure e-voting web application integrating facial recognition using OpenCV and a KNN classifier for voter authentication. • Designed a custom proof-of-work blockchain to ensure tamper-proof and double-vote-free vote recording. • Implemented live results to provide real-time election monitoring.	

CERTIFICATES & ACHIEVEMENTS

Introduction to Machine Learning	IIT Madras (NPTEL/Swayam)
• 12-week course · Jul–Oct 2025	
Marketing Analytics – Elite	IIT Kharagpur (NPTEL/Swayam)
• 12-week course · Jan–Apr 2026	
The Bits and Bytes of Computer Networking	Google, via Coursera
• Completed Nov 2025	